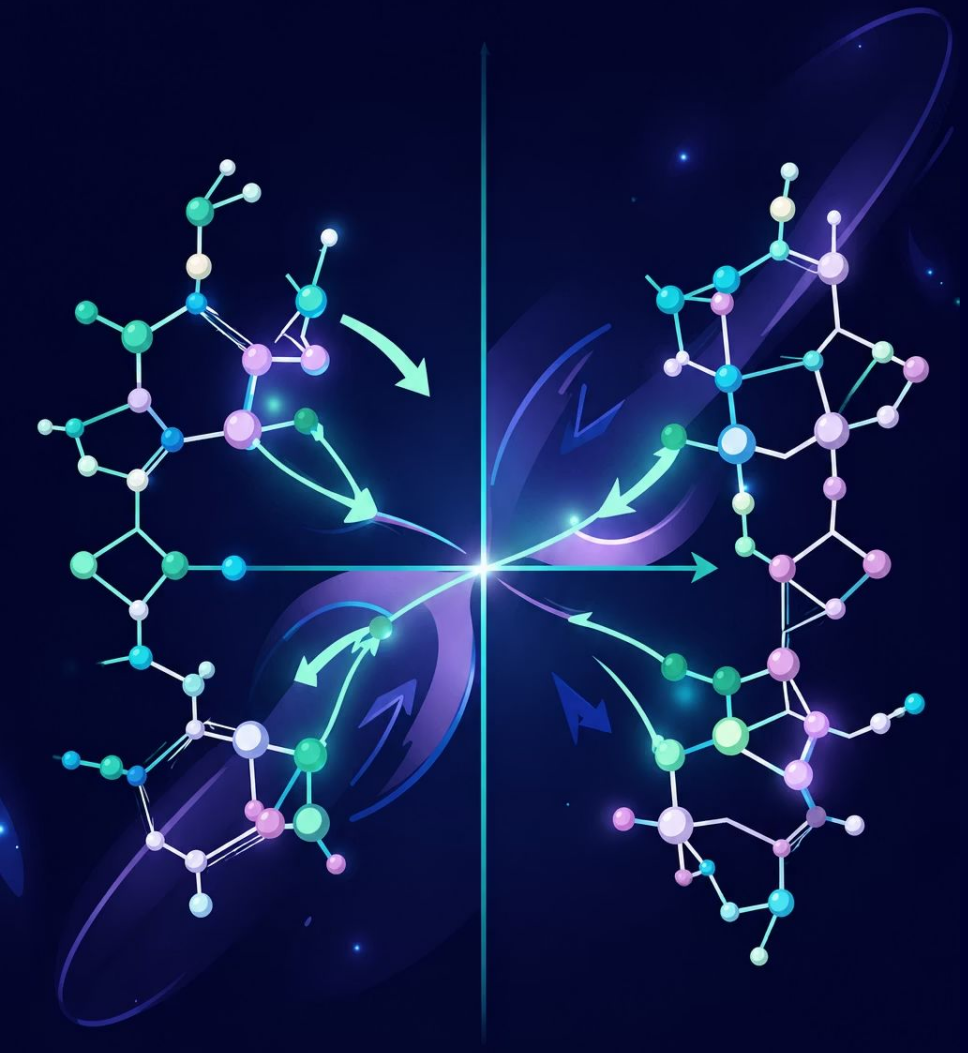


RESEARCH PRESENTATION

The Jafarpour Model

Stochastic Dynamics of a Modified Frank Model for Spontaneous Chiral Symmetry Breaking

Presented by **Saswata Chatterjee**

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Motivation

Chiral Symmetry Breaking

How symmetry between left- and right-handed states is lost spontaneously.

Studying Homochirality

We want to inspect if homochirality can be obtained in absence of explicit chiral inhibition.

Experimental Evidences

Unlike the original Frank's Model which had no experimental evidence for numerous years barring one, the modified model was observed in recent ribozyme-mediated RNA replication experiments.

Reintroduction of the Decayed Specie

The first of the two Jafarpour models examined a model where we have no chiral inhibition however there is no way for to retrieve the once Decayed species either.

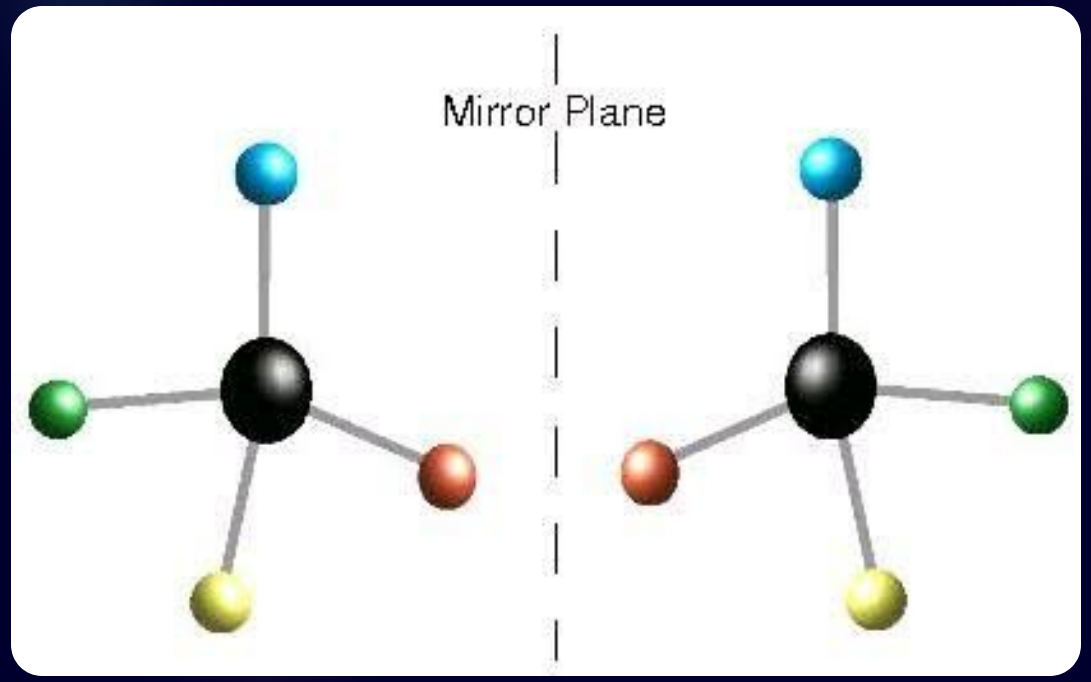
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HOMOCHIRALITY INTRODUCTION

Introduction to Homochirality:

Chirality

- A chiral molecule cannot be superimposed on its mirror image. Its mirror-image forms are called **enantiomers**.
- They are often labeled L (left-handed) and D (right-handed).
- **Amino Acids**: Nearly exclusively L-chiral.
- **Sugars**: Nearly exclusively D-chiral.



Racemic Mixture

- Without a chiral influence, reactions usually form both **enantiomers** equally.
- A 50:50 mix of L and D enantiomers is a **racemic mixture.** It has no overall chirality.



The image perfectly represents the idea of Enantiomers:
Non superimposible mirror images

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Jafarpour Reaction Model (2017)

The new Jafarpour model introduced reversible reactions as opposed to a linear decay

Reaction Model:

$$A + D \xrightarrow{k_a} 2D$$

$$A + L \xrightarrow{k_a} 2L$$

$$A \underset{k_d}{\overset{k_n}{\rightleftharpoons}} D$$

$$A \underset{k_d}{\overset{k_n}{\rightleftharpoons}} L$$

⚠ Changes introuced

- A new parameter K_n was added to make sure we can reintroduce the species back into this medium.
- From the original franks model the chiral inhibition step was completely removed.

Mean field equations:

$$\frac{dA}{dt} = -k_a A(D + L) - 2k_n A + k_d(D + L)$$

$$\frac{dD}{dt} = k_a AD + k_n A - k_d D$$

$$\frac{dL}{dt} = k_a AL + k_n A - k_d L$$

The total concentration is conserved

$$A + D + L = N$$

This ensures we can have a fixed number of molecules in our system at any given time.

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Important Parameters

1. Chiral Order Parameter (ω)

$$\boxed{\omega = \frac{d - l}{d + l}}$$

The important limiting cases for this are:

$$\omega = 0 \qquad \text{for the racemic state}$$

$$\omega = 1 \qquad \text{for the } D\text{-homochiral state}$$

$$\omega = -1 \qquad \text{for the } L\text{-homochiral state}$$

2. Rate of change

We can define

$$\Delta = D - L$$

from solving the mean field equations we can obtain a relation

$$\frac{d\Delta}{dt} = k_a A(D - L) - k_d(D - L)$$

this will be especially useful for the analysis of various regimes later on

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DETERMINISTIC ANALYSIS

Deterministic Analysis

Since there are 3 parameters, namely K_a, K_n, K_d we would want to examine their behavior and understand how this system behaves

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Case 1: We consider the modified reaction system K_a = K_n = K_d =1.0, N =1

$$A = 1 - D - L$$

We can substitute the value of A in the above equations and find the two rate equations in terms of D and L

$$\boxed{\frac{dD}{dt} = 1 - D - L - D^2 - DL}$$

$$\boxed{\frac{dL}{dt} = 1 - D - L - DL - L^2}$$

Equating both to 0

$$(L - D)(L + D) = 0$$

so we have two solutions, these are cases where we will find an equilibrium provided the nullclines of the system are present

$$\boxed{D = L}$$

$$\boxed{D = -L}$$

Since one of the two solutions is physically not feasible we will only take into account the $\boxed{D = L}$ solution for our calculations moving forward

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DETERMINISTIC ANALYSIS EQUILIBRIUM & JACOBIAN

Equilibrium Points and Jacobian Analysis

now we must find the two equilibrium points of the hence under the given condition we equate the nullcline putting the mean field equation to 0

we find two equilibrium points

$$\left(-\frac{1+\sqrt{3}}{2}, -\frac{1+\sqrt{3}}{2}\right)$$

$$\left(\frac{\sqrt{3}-1}{2}, \frac{\sqrt{3}-1}{2}\right)$$

since concentration can not be negative the negative roots are meaningless and we have only one useful solution. This point is the **RACEMIC EQUILIBRIUM** point of the system. Now we must perform a stability analysis via eigen decomposition to see how it behaves.

JACOBIAN CONSTRUCTION

Construction of the Jacobian Matrix

we can expand this into

$$J(D,L) = \begin{pmatrix} \frac{\partial f}{\partial D} & \frac{\partial f}{\partial L} \\ \frac{\partial g}{\partial D} & \frac{\partial g}{\partial L} \end{pmatrix}$$

---

we can expand this into

$$J(D,L) = \begin{pmatrix} -1-2D-L & -1-D \\ -1-D & -1-D-2L \end{pmatrix}$$

---

subsitutiting the values of D and L in the jacobian we obtain

$$J^* = \begin{pmatrix} \frac{1-3\sqrt{3}}{2} & -\frac{1+\sqrt{3}}{2} \\ -\frac{1+\sqrt{3}}{2} & \frac{1-3\sqrt{3}}{2} \end{pmatrix}$$

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Now that we have obtained the jacobian matrix at the desired equilibrium point we can now perform the stability test

$$\begin{vmatrix} \frac{1-3\sqrt{3}}{2} - \lambda & -\frac{1+\sqrt{3}}{2} \\ -\frac{1+\sqrt{3}}{2} & \frac{1-3\sqrt{3}}{2} - \lambda \end{vmatrix} = 0$$

solving this determinant yields two eigen values

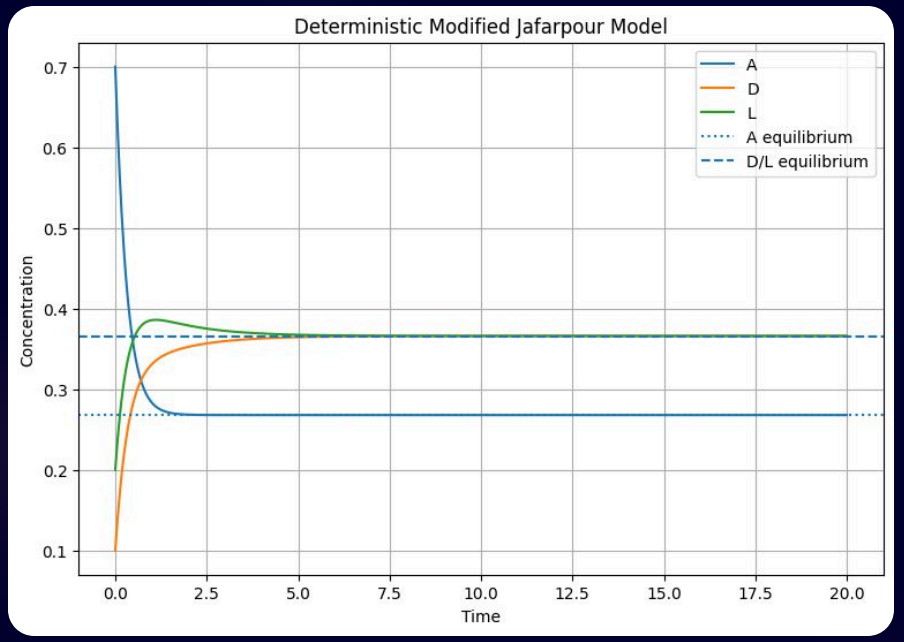
Eigenvalue 1

$$\lambda_1 = -2\sqrt{3}$$

Eigenvalue 2

$$\lambda_2 = 1 - \sqrt{3}$$

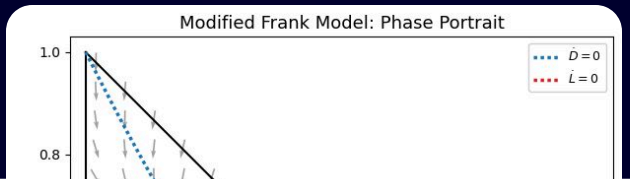
since all eigen values are negative we can safely say that this point would be **STABLE** and attract all trajectories.



Final concentrations:
A = 0.267949
D = 0.366025
L = 0.366025

Theoretical equilibrium:
A* = 0.267949
D* = 0.366025
L* = 0.366025

Phase Space Analysis



Conclusion:

A very important thing to observe here is unlike the previous models this has the ability to reintroduce
into the system, essentially if the concentration of a specie is higher it decays more, if its lesser it decay
have an equal chance of being produced from the achiral precursor as it has no dependance on the D or L concentration. More

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Case 2:

We will now examine a case where $K_a = 1.0$, K_n =1.0 and K_d =0.0001 so essentially a lot of preference towards the autocatalysis

---

Now the modified mean nullcline equations are:

$$\boxed{\frac{dD}{dt} = 1 - 1.0001D - L - D^2 - DL = 0}$$

$$\boxed{\frac{dL}{dt} = 1 - D - 1.0001L - DL - L^2 = 0}$$

Equating both of them to 0 simultaneously again we obtain

$$\boxed{(L - D)(D + L + 0.0001) = 0}$$

Hence we have 2 different solutions to this

$$\boxed{D = L}$$

$$\boxed{D = -0.0001 - L}$$

Since one of them is physically impossible $D = L$ is the only feasible solution to this system

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Stability analysis

We can use the condition to solve the nullclines, and after plugging the value we see that there is only one feasible solution at equilibrium

$$(A^*, D^*, L^*) \approx (0.00003333,\ 0.4999833,\ 0.4999833)$$

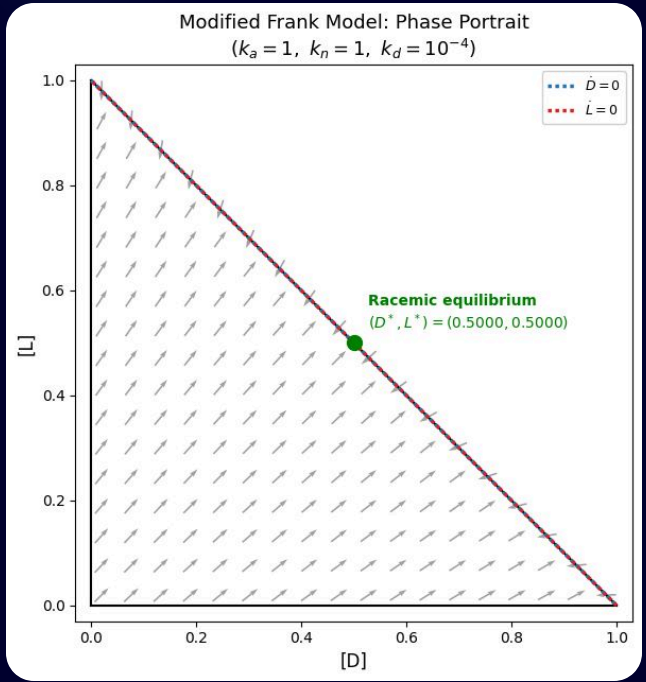
now after performing the stability test we for the 2D jacobian matrix wrt D and L we can observe that the eigen values are

$$\lambda_1 \approx -3.00003333$$

$$\lambda_2 \approx -6.66674 \times 10^{-5}$$

Since both eigen values are negative we have a stable fixed point here, as can be shown from the phase space diagram.

Phase space analysis:



Analysis:

- For this particular system the rate of change stands as:

$$\frac{d\Delta}{dt} = (0.9999 - D - L)(D - L).$$

in most scenarios This could lead to a case of homochiral amplification however there would come a time when eventually the first term is more dominant and negative.

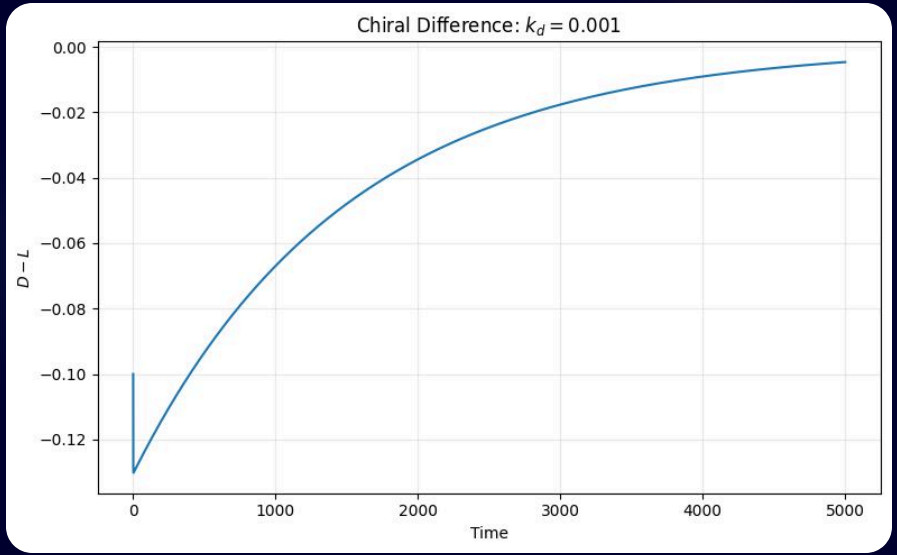
- The chiral difference here initially amplifies since the $D - L$ term is positive hence keeps increasing the rate unless the negative term is dominant. This is better observed in the ODE simulations.

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DETERMINISTIC SIMULATIONS VISUAL SUMMARY

Determinisitc Simulations:

Simulation describing the difference between the two species



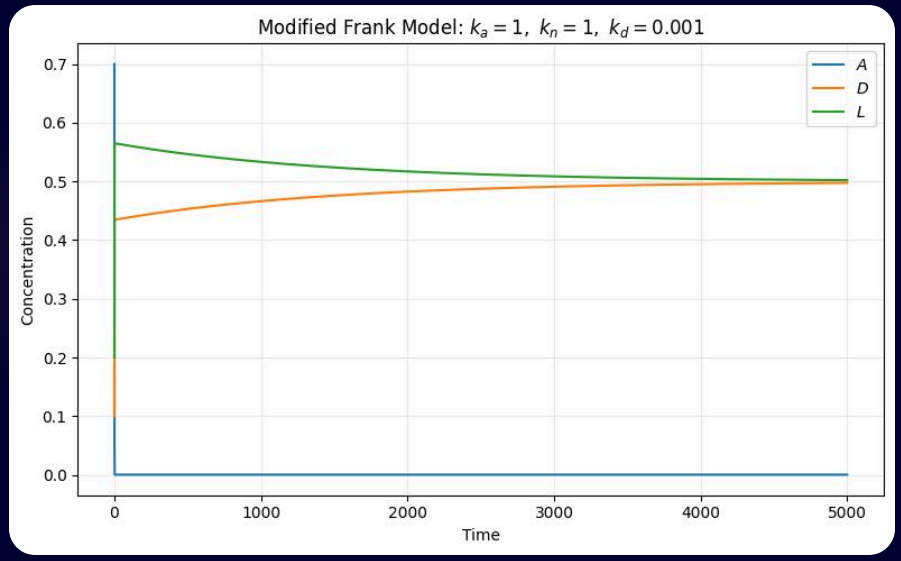
Inference:

This is a very useful observation and inference and we will look into the stochastic simulation of this case in more a detail later on.

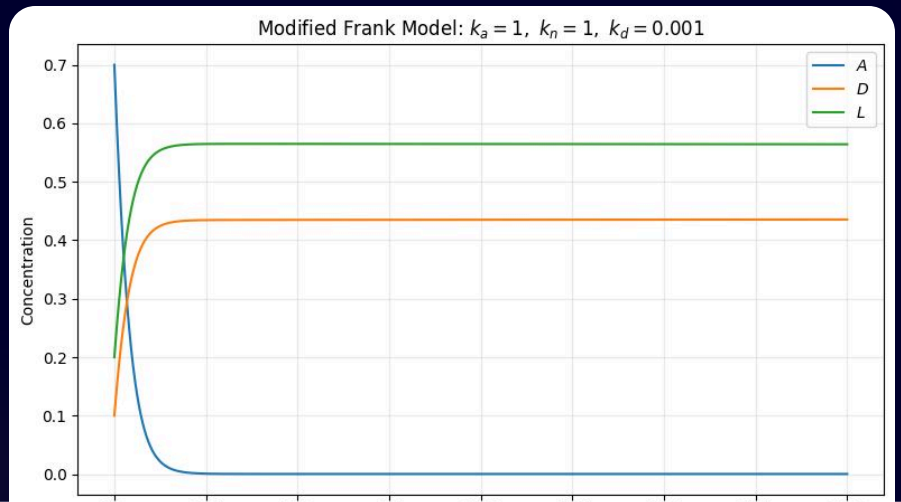
The deterministic simulation gives us an interesting picture where noise induced simulations could give us homochirality.

We will now analyse how the parameter K_n has effect on the system deterministically. However before we analyse all possible scenarios we would look at what the parameter is even responsible for in the first place. This will allow us to have a better understanding before going into the mathematical treatment.

Iniital conditions: $A_0 = 0.70 \;\; D_0 = 0.10 \; L_0 = 0.20$



If we look at at much smaller time scale:



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DETERMINISTIC SIMULATIONS VISUAL SUMMARY

Case 3 (Study of the parameter K_n)

If we analyse the mean field equations we can clearly see that K_n appears **identically in both D and L equations**. Therefore, it cannot directly create a D/L preference

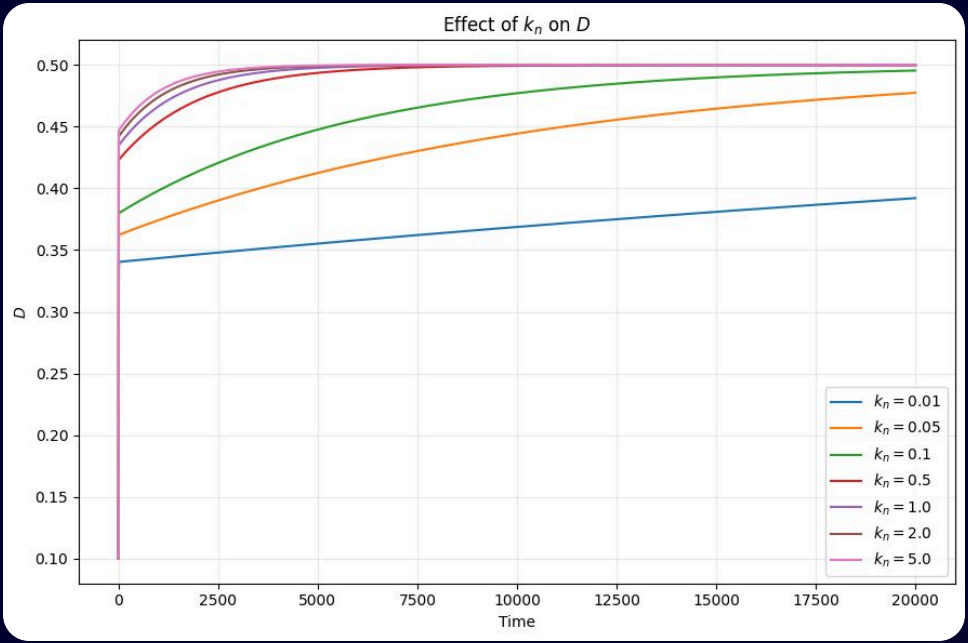
$$\boxed{\frac{dD}{dt} = k_a AD + k_n A - k_d D} \qquad \boxed{\frac{dL}{dt} = k_a AL + k_n A - k_d L}$$

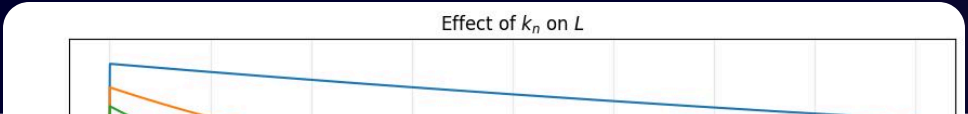
hence it doesnt directly contribute to either suppression or amplification of homochirality however, it equally contributes to the formation of each of the species with no predetermined bias. If anything this could be a factor against the homochirality.

Deterministic Analysis:

Iniital conditions: $A_0 = 0.70$ $D_0 = 0.10$ $L_0 = 0.20$

K_a = 1.0 and K_d = 10^{-4}





Inference:

- The deterministic simulations makes sense because a larger k_n rapidly converts Ainto D and L **symmetrically**.

So the spontaneous pathway is effectively saying:

$$A \xrightarrow{k_n} \begin{cases} D \\ L \end{cases}$$

- We can see that K_n has no bearings on the final state of the system but only on the rate at which they attain the equilibrium.
- We can say that K_n is a driving force towards racemic mixture hence increasing it will only allow us to reach the stable equilibrium faster.

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STABILITY ANALYSIS EQUILIBRIUM EIGENVALUES

Stability Analysis:

Let us consier a general case scenario for the system to be at equilibrium the two possible conditions after solving the nullcline are:

$$\boxed{D = L}$$

$$\boxed{A = \frac{k_d}{k_a}}$$

However the second solution isn't particularly feasible since substituting it back into the original equation of D' gives us

$\boxed{k_d D + k_n A - k_d D = k_n A = 0}$ which is an impossible criteria since both the A and K_n >0 hence $\boxed{D = L}$ is the only criteria that is satisfied.

After constructing the Jacobian matrix:

$$\boxed{J(D,L) = \begin{pmatrix} k_a(A-D) - k_n - k_d & -k_a D - k_n \\ -k_a L - k_n & k_a(A-L) - k_n - k_d \end{pmatrix}}$$

We can apply the equilibrium critera of $D^* = L^* = x$ and hence $A = N - 2x$ hence reconstruct the jacobian:

$$\boxed{J^* = \begin{pmatrix} k_a(A^* - x) - k_n - k_d & -k_a x - k_n \\ -k_a x - k_n & k_a(A^* - x) - k_n - k_d \end{pmatrix}}$$

From this we can perform a characteristic equation and obtain our eigen values:

$$\lambda_1 = k_a(A^* - x) - k_n - k_d - k_a x - k_n$$

which can be further simplified as:

$$\boxed{\lambda_1 = -2k_a x - 2k_n - \frac{k_n A^*}{x}}$$

$$\lambda_2 = k_a(A^* - x) - k_n - k_d + k_a x + k_n$$

using the condition of equilibrium this turns out as:

$$\lambda_2 = k_a A^* - k_d$$

or more conveniently

$$\boxed{\lambda_2 = -\frac{k_n A^*}{x}}$$

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STABILITY ANALYSIS EIGENVECTOR MODES RELAXATION TIME

Eigen vector modes and Relaxation time

At the racemic equilibrium, the Jacobian is

$$J^* = \begin{pmatrix} k_a(A^* - x) - k_n - k_d & -k_a x - k_n \\ -k_a x - k_n & k_a(A^* - x) - k_n - k_d \end{pmatrix}$$

Symmetric and Antisymmetric Modes

Symmetric mode: Eigenvector D and L change in the same direction, changing the total chiral concentration.

$$\lambda_{\text{sym}} = k_a(N - 4x) - 2k_n - k_d \qquad v_{\text{sym}} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Antisymmetric mode: Eigenvector D and L change in opposite directions, changing the chiral imbalance while keeping constant.

$$\lambda_{\Delta} = k_a(N - 2x) - k_d \qquad v_{\Delta} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Relaxation Time

Near equilibrium, perturbations evolve as

$$\delta(t) = \delta(0)e^{\lambda t}$$

The characteristic relaxation time is

$$\tau = \frac{1}{|\lambda|}$$

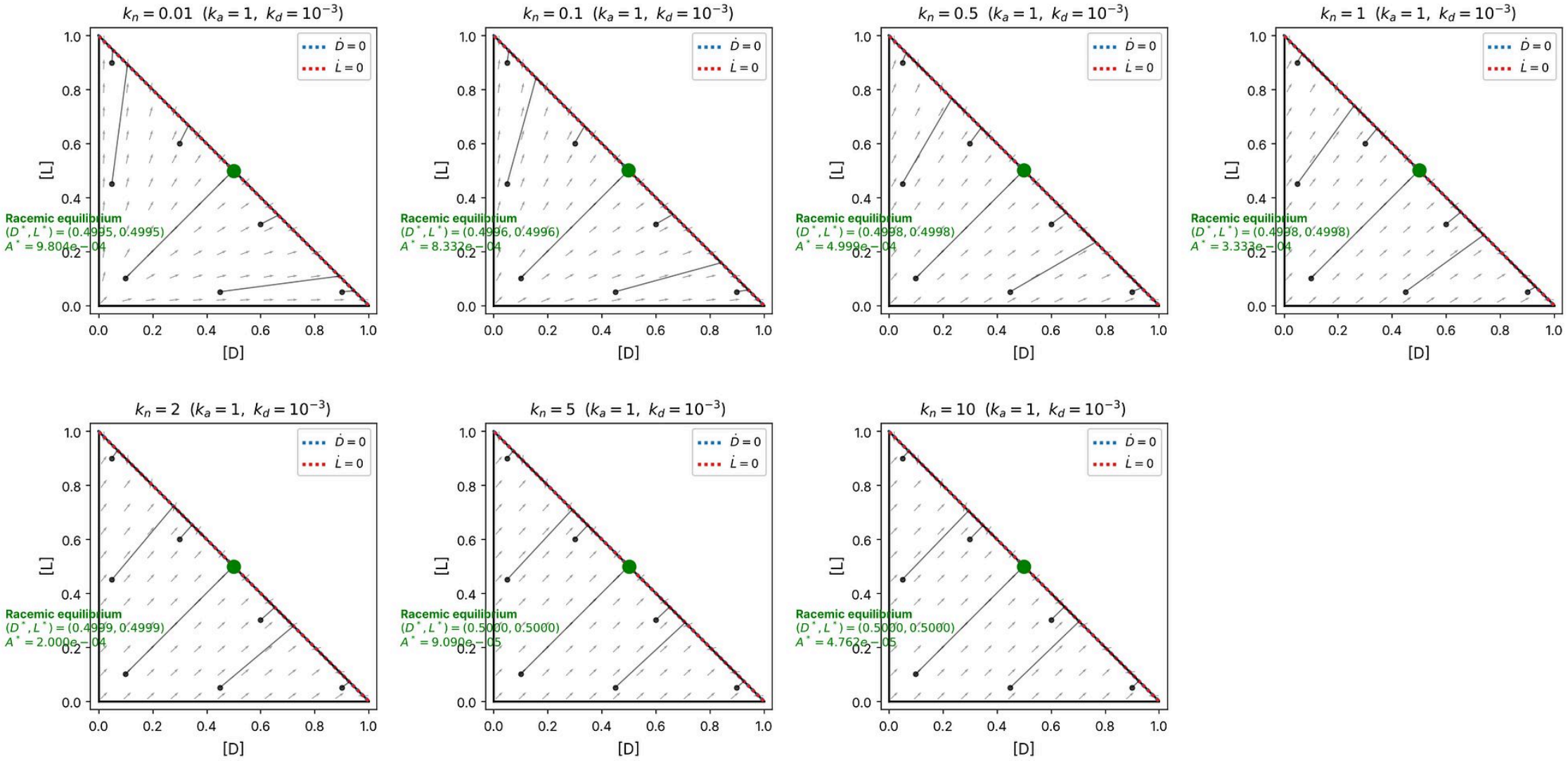
Thus,

$$\tau_{\text{sym}} = \frac{1}{|\lambda_{\text{sym}}|}, \qquad \tau_{\Delta} = \frac{1}{|\lambda_{\Delta}|}$$

- Both eigenvalues are negative, indicating local stability. The antisymmetric mode can relax much more slowly because its eigenvalue is closer to zero.

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Phase portraits:



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ANALYSIS PHASE BEHAVIOR

Summary:

k_n	λ_{sym}	τ_{sym}	λ_{Δ}	τ_{Δ}
0.01	−1.01904	0.9813	-1.9627×10^{-5}	50,951
0.1	−1.19933	0.8338	-1.6678×10^{-4}	5,996
0.5	−2.00000	0.5000	-5.0012×10^{-4}	1,999.5
1	−3.00033	0.3333	-6.6674×10^{-4}	1,499.8
2	−5.00060	0.2000	-8.0003×10^{-4}	1,250
5	−11.00082	0.0909	-9.0910×10^{-4}	1,100
10	−21.00091	0.0476	-9.5238×10^{-4}	1,050

CONCLUSION

- Symmetric mode: Describes changes in the total chiral concentration, $D + L$, and decays rapidly due to its short relaxation time.
- Antisymmetric mode: Describes changes in the chiral imbalance, $D + L$, and decays slowly due to its much longer relaxation time.
- Trajectory behaviour: Trajectories approach the region near the hypotenuse relatively quickly, then move slowly along it toward the racemic equilibrium.
- Effect of K_n : At low K_n , the antisymmetric relaxation time is particularly long, so trajectories may not reach the green equilibrium point within the simulation time of 400 units.

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ANALYSIS | STOCHASTIC MODEL | LANGEVIN EQUATION

Stochastic Model

Langevin Equation

The stochastic reduction separates the dynamics into a drift term that pulls the system toward racemic balance and a noise term that captures finite-size fluctuations.

Derivation of the Drift Term

Define the total chiral concentration and concentration difference:

$$r = d + l, \qquad \Delta = d - l, \qquad \omega = \frac{\Delta}{r}$$

Subtracting the mean-field equations for D and L gives

$$\frac{d\Delta}{dt} = (k_a a - k_d)\Delta.$$

Adding the equations and using conservation, A=N-r gives

$$\frac{dr}{dt} = -k_d r + 2k_n(n - r) + k_a(n - r)r.$$

Applying the quotient rule and simplifying,

$$\frac{d\omega}{dt} = -2k_n \frac{n - r}{r}\omega.$$

At steady state, $dr/dt = 0$

$$\frac{n - r^*}{r^*} = \frac{k_d}{k_a r^* + 2k_n}.$$

Therefore, the drift term is

$$A_\omega = -\frac{2k_n k_d}{k_a r^* + 2k_n}\omega.$$

Under the approximations $(r^* \approx n)$ and $k_a n \gg 2k_n$:

$$A_\omega \approx -\frac{2k_n k_d V}{N k_a}\omega.$$

Derivation of the Noise Term

The diffusion matrix describes fluctuations in the original variables A,D,L

Define the chirality variable

$$\omega = \frac{d - l}{d + l}, \qquad r = d + l$$

Its gradient is

$$\nabla\omega = \begin{pmatrix} 0 \\ \frac{1-\omega}{r} \\ -\frac{1+\omega}{r} \end{pmatrix}$$

Transform the diffusion matrix to obtain the diffusion coefficient for chirality:

$$B_{\omega\omega} = \nabla\omega^T \mathbf{B} \nabla\omega.$$

Using the diffusion matrix elements,

$$B_{\omega\omega} = \epsilon\left[\frac{k_d + a k_a}{r}(1 - \omega^2) + \frac{2a k_n}{r^2}(1 + \omega^2)\right].$$

The noise amplitude is the square root of the diffusion coefficient:

$$\sigma_\omega = \sqrt{B_{\omega\omega}}.$$

Hence, the effective stochastic equation is

$$\frac{d\omega}{dt} = A_\omega + \sqrt{B_{\omega\omega}}, \eta(t)$$

where $\eta(t)$ is Gaussian white noise satisfying

$$\langle \eta(t) \rangle = 0, \qquad \langle \eta(t)\eta(t') \rangle = \delta(t - t').$$

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ANALYSIS | LANGEVIN EQUATION | STATIONARY DISTRIBUTION

Analysis of the Langevin equation

Langevin equation

$$\frac{d\omega}{dt} = -\frac{2k_n k_d V}{N k_a}\omega + \sqrt{\frac{2k_d}{N}(1-\omega^2)}\,\eta(t)$$

The Langevin equation for our system stands as

Stationary probability distribution

For this equation, if we look at the stationary probability distribution, we can obtain a dimensionless parameter α which directly the distribution of the particle at the final position:

- The stationary probability distribution is given as:

$$P_s(\omega) = \frac{\Gamma\left(\alpha+\frac{1}{2}\right)}{\sqrt{\pi}\Gamma(\alpha)}(1-\omega^2)^{\alpha-1}, \qquad -1 \le \omega \le 1$$

where

$$\boxed{\alpha = \frac{k_n V}{k_a}}$$

This is arguably the most important parameter here,

Absence of K_d

- increases the magnitude of the restoring drift.
- K_d also increases the noise amplitude.
- In the simplified equation, these effects scale in a way that makes K_d cancel from the stationary distribution's shape.

How α scales

- The gamma prefactor is the normalization constant here it is insiginifcant to the distribution.
- $0 < \alpha < 1$: the distribution is enhanced near $\omega = \pm 1$, favoring highly homochiral states.
- $\alpha = 1$: the distribution is uniform.
- $\alpha > 1$: the distribution is concentrated around $\omega = 0$, favoring racemic state

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ANALYSIS GILLESPIE SIMULATIONS HOMOCHIRALITY

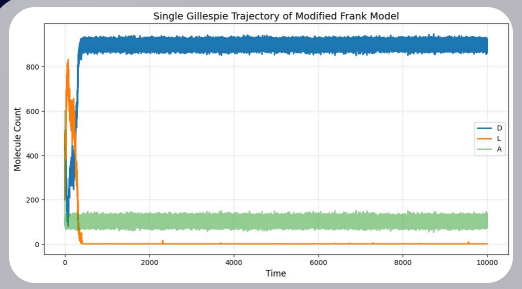
Gillespie Simulations:

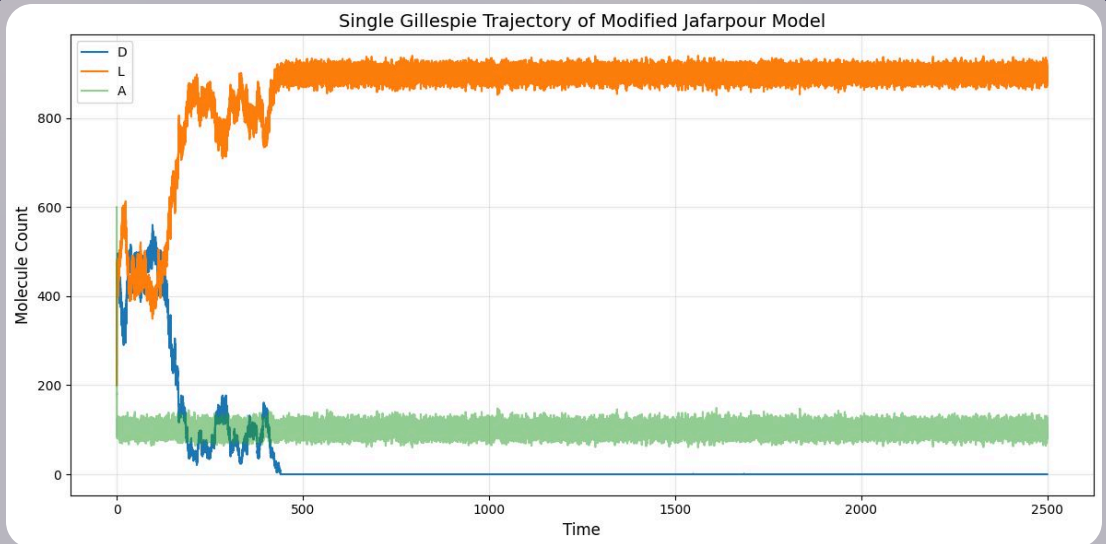
Since the ensemble trajectories are more likely to yield results closely resembling a deterministic solution its better if we first look at singular trajectories. Having a lower value of α seems to favour a homorchiral state from our theoretical calculations so we must see how it matches upto our simulations:

Case 1:

Conditions → $K_a = 1.0$, $K_d = 1.0$, $K_n = 10^{-4}$, Volume=100 Units, α =0.01, A_0 = 600, $L_0 = D_0$ =200

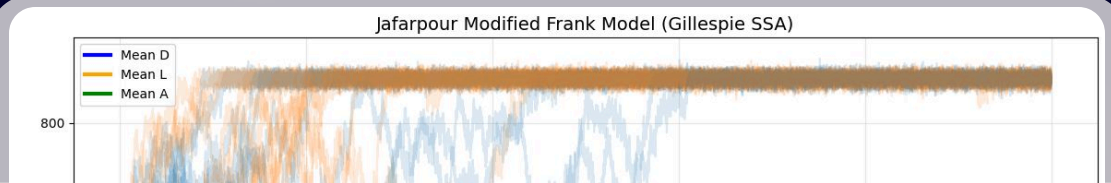
Theoretically this edge case has a very probability of attaining homochirality

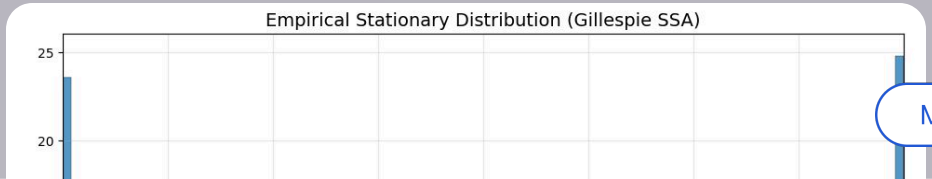




We are indeed able to attain a homochiral state which perfectly matches our theoretical calculations

If we analyse an ensemble trjectory:



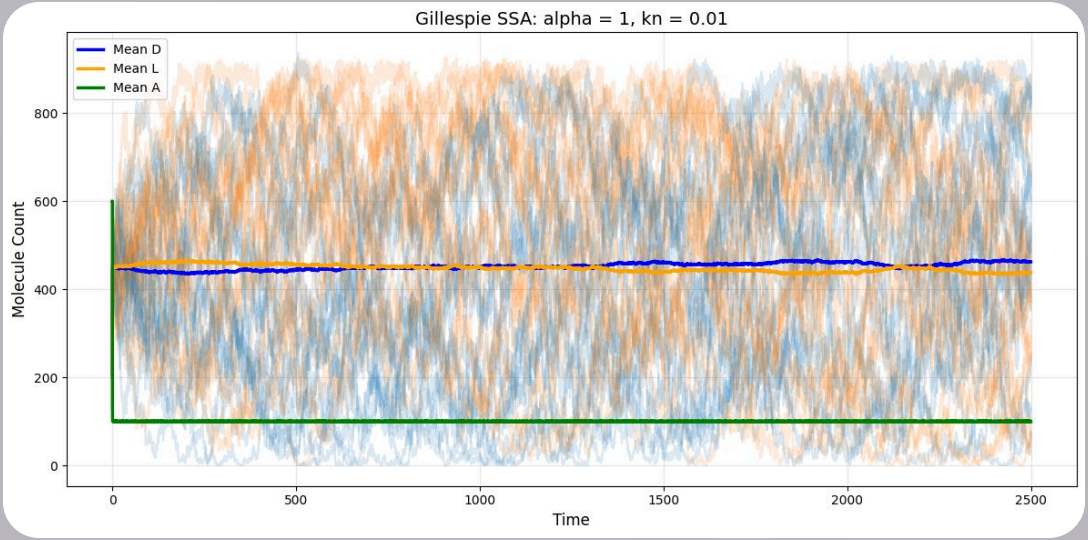


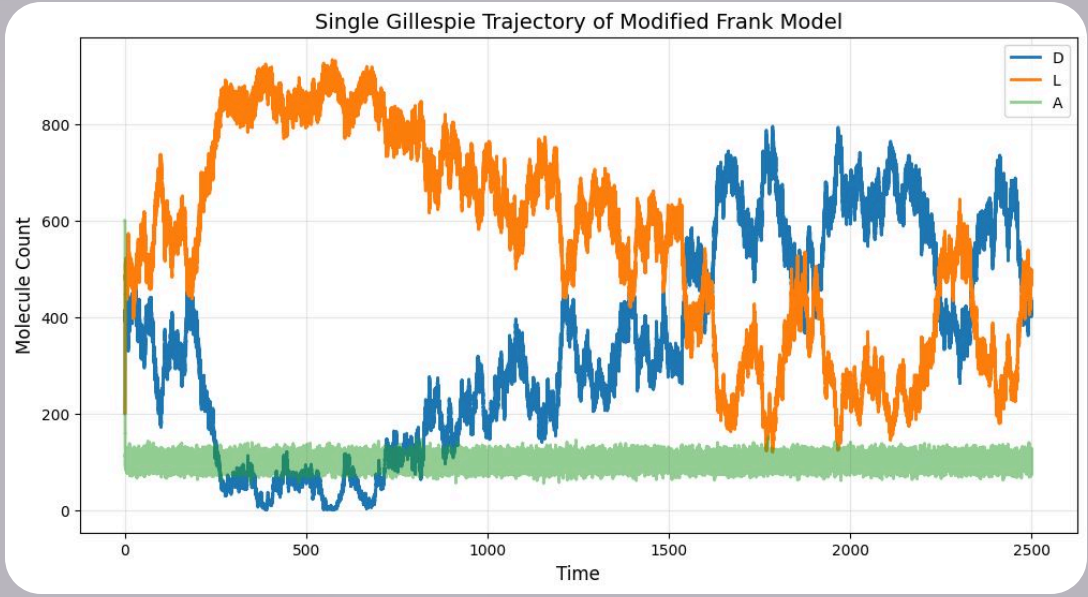
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GILLESPIE SIMULATIONS | CASE 2

Case 2:

Conditions→$K_A = 1.0$, $K_d = 1.0$, $K_n = 10^{-2}$, Volume=100 Units, α =1.0, A_0 = 600, $L_0 = D_0$ =200





As we can see from the simulation at slightly higher regimes of K_n values or α values we actually fail to arrive at a state of homochirality and the mixture prefers to be racemic overall. The stationary distribution is very useful to understand this as the homochiral tendency scales down with increase of α values

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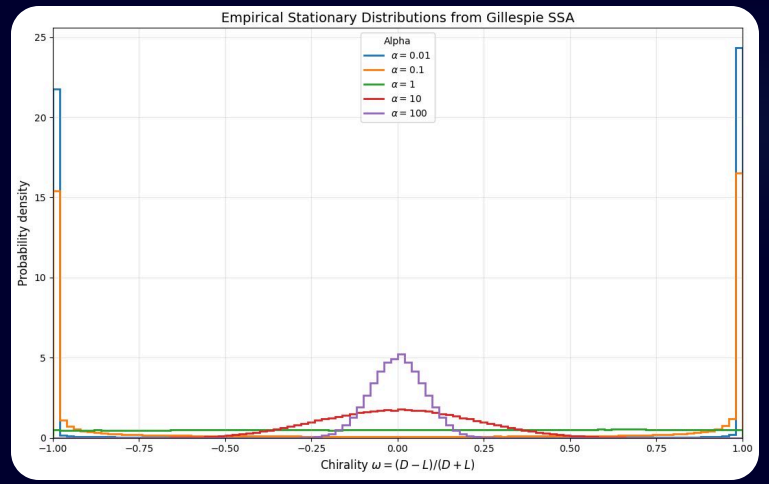
STATIONARY DISTRIBUTION | RESULTS | INFERENCE

Stationary distribution over various parameters:

(Consider the Volume to be 100 units across all of the simulations)

alpha	kn	Mean A	Mean D	Mean L	D-L	Mean omega
0.01	0.0001	100.310	469.735	429.955	39.780	0.05270
0.1	0.001	100.282	464.374	435.344	29.030	0.02480
1	0.01	99.617	462.919	437.464	25.455	0.01884
10	0.1	98.334	455.069	446.597	8.472	0.00084
100	1	82.141	458.616	459.243	-0.627	0.00001

Stationary Distribution



Inference

- $\alpha = 0.01$: The distribution is strongly concentrated near $\omega = \pm 1$, indicating a preference for near-homochiral states.
- $\alpha = 0.1$: The distribution remains enhanced near both endpoints, although the endpoint peaks are lower than for $\alpha = 0.01$.
- $\alpha = 1$: The distribution is approximately uniform across the chirality range, with no strong preference for racemic or homochiral states.
- $\alpha = 10$: The distribution becomes concentrated around $\omega = 0$, indicating a preference for states with similar D and L populations.
- $\alpha = 100$: The distribution is even more sharply peaked around $\omega = 0$, indicating a strong preference for racemic states.

- ***Overall trend***: As $\alpha = k_n V / k_a$ increases, the distribution shifts from endpoint-enhanced states toward the centre, suggesting that increasing spontaneous production relative to autocatalysis favors reduced chiral imbalance in the reduced Langevin model.

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CONCLUSION SUMMARY KEY TAKEAWAYS

Conclusion:

- The modified model provides a more flexible framework for studying chiral symmetry breaking by incorporating spontaneous production, autocatalysis, and reverse reactions.
- It captures stochastic fluctuations, allowing us to observe how random events can lead to temporary dominance of one enantiomer.
- The model overall does a great job in establishing Homochirality albeit temporary but the fact that it has more experimental evidence makes for a strong case for it to be useful in the future studies.
- The results depend on simulation time, system size, initial conditions, and the number of trajectories considered. It is heavily reliant on the parameters and perhaps more strict in terms of the regimes which allow for homochirality to exist as compared to Franks Model which has an inherent amplification.
- The modified model offers a useful way to investigate the relationship between stochastic fluctuations, reaction kinetics, and chiral symmetry breaking. Its main advantage is that it captures the dynamic nature of chirality, while its main limitation is that near-homochirality does not necessarily imply permanent or complete homochirality.

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REFERENCES ACKNOWLEDGEMET

References and Acknowledgement:

- Frank, F. C. (1953)

Frank, F. C. On spontaneous asymmetric synthesis. Biochimica et Biophysica Acta, 11, 459–463.

- Jafarpour et al.

Jafarpour, F., Biancalani, T., & Goldenfeld, N. (2015). Noise-induced mechanism for biological homochirality. Physical Review Letters.

- Soai Reaction

Soai, K., Shibata, T., Morioka, H., & Choji, K. (1995). Asymmetric autocatalysis and amplification of enantiomeric excess. Nature.

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Order Emerges

Homochirality reminds us that nature can choose one handedness, turning symmetry into life’s essential asymmetry.

> “From mirrored possibilities, one elegant form prevails.”

THANK YOU!!



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